

SYED AHMED HASSAN SHAH

MPhil Researcher, Central South University | Former AI Engineer, NCAI (Pakistan) & NeuroCare.AI (USA)

Top 1% MPhil Cohort (95.75%) | CSC Scholar | 10+ Q1 Peer-Reviewed Publications

Co-author of the book Smart Infrastructure Management (Elsevier)

International Students Dormitory, Railway Campus, Central South University, Changsha, Hunan, China

Email | Google Scholar | LinkedIn | Personal Website | ResearchGate | GitHub

RESEARCH PROFILE

I am currently pursuing an MPhil in Civil Engineering at Central South University, China, where my research focuses on intelligent infrastructure using computer vision, 3D perception, multimodal learning, generative AI, and predictive digital-twin approaches for railway inspection, component-condition assessment, and smart monitoring. This work builds on my earlier training in Computer Science and professional experience in AI and computer vision. I previously worked at the National Center of Artificial Intelligence (NCAI), Pakistan, on government-funded medical AI projects involving MRI, CT, X-ray, and ultrasound-based diagnostic imaging, and later at NeuroCare.AI, USA, on medical computer-vision and AI-assisted clinical applications. My research has led to multiple peer-reviewed publications, book chapters, and contributions to an academic book on smart infrastructure management, alongside collaborations across civil engineering, geotechnical AI, and smart transportation systems.

RESEARCH FOCUS

- Computer Vision & 3D Perception
- Multimodal & Generative AI
- Digital Twins & Predictive Analytics
- Smart Infrastructure Monitoring
- Railway Inspection & Condition Assessment

CURRENT DIRECTION

Trustworthy multimodal AI for robust perception, monitoring, and decision support in safety-critical infrastructure.

APPLICATION DOMAINS

Rail Infrastructure | Civil Engineering
Geotechnical AI | Smart Transportation

EDUCATION

Master of Philosophy in Civil Engineering (MPhil)

Central South University (CSU), Changsha, China

Dates: Sep 2024 - Present **CGPA: 3.83 / 4.00** **Top 1% of Class**

Thesis: Rail Fastener Alignment & Fault Detection for On-Device Surveillance in Low-Visibility Conditions Using Multimodal Generative AI.

Research Focus: Computer Vision, 3D Perception, Multimodal Learning, Generative AI, Intelligent Railway Inspection, Predictive Digital Twins.

Bachelor of Science in Computer Science (BSCS)

COMSATS University Islamabad, Attock, Pakistan

Dates: Jul 2018 - Jul 2022 **CGPA: 3.24 / 4.00** **Springer LNNS**

Thesis: Transfer Learning-Based Contagious Diseases Transmission Video Surveillance: A COVID-19 Case Study.

Highlight: Published as a Springer LNNS book chapter and presented at ISSET.

RESEARCH & PROFESSIONAL EXPERIENCE

CSU

Research Assistant - Smart Infrastructure & AI

Sep 2024 - Dec 2025

Central South University, China Smart Infrastructure Management

- Developed multimodal railway-inspection pipelines integrating computer vision, 3D point clouds, generative enhancement, and deep learning for fastener detection, alignment, and condition assessment.
- Advanced explainable and on-device maintenance intelligence, contributing to research on GNN-based tightness assessment, digital twins, low-light AIoT, and infrastructure defect analysis.

NC.AI

AI Engineer - Medical Computer Vision

May 2024 - Sep 2024

NeuroCare.AI, United States (Remote)

- Developed computer-vision methods for neuroimaging and brain-hemorrhage analysis, translating research models into practical AI-assisted clinical decision-support workflows.

NCAI

Computer Vision Researcher

Sep 2022 - Sep 2023

National Center of Artificial Intelligence (NCAI), Pakistan Medical Imaging & Diagnostics Lab

- Contributed to government-funded medical-AI projects across MRI, CT, X-ray, and ultrasound, developing detection, segmentation, and classification pipelines for tuberculosis, breast cancer, and brain tumors.

PUBLICATIONS & SCHOLARLY OUTPUT

Intelligent Infrastructure & Civil AI

S. Qiu, **S. M. A. H. Shah**, W. Wang, et al.. "**AIoT-Based Rail Fastener Assessment in Low-Light Environment Using Focal Modulated Unsupervised Image Enhancement and Generative Segmentation**." **Journal of Transportation Engineering, Part A: Systems**, ASCE, 2026.

Q. Zaheer, **S. M. A. H. Shah**, W. Wang, et al.. "**Multimodal Graph Neural Network Framework for Railway Fastener Tightness Assessment from High-Resolution Point Clouds**." **Engineering Applications of Artificial Intelligence**, Elsevier, 2026.

S. Qiu, Q. Zaheer, **S. M. A. H. Shah**, et al.. "**Vector-Quantized Variational Teacher and Multimodal Collaborative Student for Crack Segmentation via Knowledge Distillation**." **Journal of Computing in Civil Engineering**, ASCE, 2025.

S. Qiu, Q. Zaheer, **S. M. A. H. Shah**, et al.. "**LiDAR-Simulated Multimodal and Self-Supervised Contrastive Digital Twin Approach for Probabilistic Point Cloud Generation of Rail Fasteners**." **Journal of Computing in Civil Engineering**, ASCE, 2025.

Q. Zaheer, S. Qiu, **S. M. A. H. Shah**, et al.. "**Intelligent Multitasking Framework for Boundary-Preserving Semantic Segmentation, Width Estimation, and Propagation Modeling of Concrete Cracks**." **Journal of Infrastructure Systems**, ASCE, 2025.

S. Qiu, Q. Zaheer, **S. M. A. H. Shah**, et al.. "**Multimodal Geometric AutoEncoder (MGAE) for Rail Fasteners Tightness Evaluation with Point Clouds & Monocular Depth Fusion**." **Measurement**, Elsevier, 2024.

Artificial Intelligence & Computer Vision

S. M. A. H. Shah, A. Rizwan, M. Sardaraz, et al.. "**Multimodal Cross-Domain Contrastive Learning: A Self-Supervised Generative and Geometric Framework for Visual Perception**." **Information Sciences**, Elsevier, 2025.

S. M. A. H. Shah, M. Q. Khan, A. Rizwan, et al.. "**Computer-Aided Diagnosis of Alzheimer's Disease and Neurocognitive Disorders with Multimodal Bi-Vision Transformer (BiViT)**." **Pattern Analysis and Applications**, Springer Nature, 2024.

S. M. A. H. Shah, M. Q. Khan, Y. Y. Ghadi, et al.. "**A Hybrid Neuro-Fuzzy Approach for Heterogeneous Patch Encoding in ViTs Using Contrastive Embeddings and Deep Knowledge Dispersion**." **IEEE Access**, IEEE, 2023.

S. M. A. H. Shah, S. F. H. Shah, A. Ullah, et al.. "**Arabic Sentiment Analysis and Sarcasm Detection Using Probabilistic Projections-Based Variational Switch Transformer**." **IEEE Access**, IEEE, 2023.

S. M. A. H. Shah, A. Rizwan, G. Atteia, M. Alabdulhafith. "**CADFU for Dermatologists: A Novel Chronic Wounds & Ulcers Diagnosis System with DHuNeT and YOLOv8**." **Healthcare**, MDPI, 2023.

Books & Book Contributions

S. Qiu, Q. Zaheer, J. Wang, C. Ai, **S. M. A. H. Shah**, et al.. "**Smart Infrastructure Management**." **Academic Book**, Elsevier, 2025.

Additional authored/co-authored book chapters include research in AI, medical imaging, computer vision, and smart systems.

SELECTED RESEARCH PROJECTS

Railway Fastener AI Inspection System 2025 – Present

Methods: Computer Vision, Generative AI, Explainable AI, Edge Deployment. Developing fault detection, health scoring, and explainable maintenance assessment for railway fasteners under challenging visual conditions.

Multimodal Rail Condition Assessment 2024 – Present

Methods: Point Clouds, Monocular Depth, GNNs, Self-Supervised Learning. Developed multimodal representations integrating geometric and visual information for component-condition assessment and digital-twin modelling.

AI-Assisted Medical Diagnostics 2022 – 2023

Methods: Medical Imaging, Deep Learning, Computer Vision. Led/contributed to AI diagnostic systems for chest, breast, and brain imaging across multiple clinical modalities under HEC/NCAI-supported projects.

HONORS & ACADEMIC DISTINCTIONS

- **Top 1% Academic Standing**, MPhil Civil Engineering, Central South University.
- **AI Workshop Instructor**, COMSATS University Islamabad / NCAI Medical Imaging & Diagnostics Lab - conducted a five-day hands-on training program on practical AI development (2023).
- **International Training on Clean Technologies for Pb and Zn Resource Utilization**, Central South University (2024).

TECHNICAL EXPERTISE

Artificial Intelligence Deep Learning, Self-Supervised Learning, Contrastive Learning, Generative AI, Knowledge Distillation, Graph Neural Networks

Computer Vision Image Segmentation, Object Detection, 3D Vision, Point Clouds, Depth Estimation, Multimodal Fusion, Explainable Vision

Scientific / Infrastructure AI Digital Twins, Intelligent Inspection, Predictive Monitoring, AIoT, Railway/Infrastructure Perception

Programming & Frameworks Python, PyTorch, TensorFlow, OpenCV, NumPy, SciPy

Additional SQL, REST APIs, IoT Systems, Research Software Development

RESEARCH COLLABORATION

Collaborated with researchers across civil engineering, geotechnical AI, smart transportation systems, medical imaging, and computer science to develop efficient AI-driven

solutions for real-world engineering and healthcare applications. Current interdisciplinary work spans intelligent infrastructure monitoring, multimodal perception, structural/civil AI, and smart transportation.

PRESENTATIONS & PROFESSIONAL ACTIVITIES

AI Healthcare Summit 2023

Jan 2023

FAST-NUCES, Islamabad, Pakistan — Presenter / Research Team

Presented NCAI medical-imaging research on breast-cancer and brain-tumor detection in collaboration with the Medical Imaging & Diagnostics Lab.

Breast Cancer Symposium 2022

Sep 2022

COMSATS University Islamabad, Pakistan — Organizing Team

Organized an industry-academic event highlighting AI-driven advances in medical imaging and cancer diagnostics.

PROFESSIONAL MEMBERSHIP

American Society of Civil Engineers (ASCE) - Transportation & Development Institute.

LANGUAGES & TESTS

Urdu	Native
English	Professional proficiency; IELTS Academic 6.5 overall (L 7.5, R 6.5, S 6.5, W 6.0)
Chinese	Intermediate conversational proficiency

ACADEMIC LINKS

Google Scholar ResearchGate LinkedIn